

2 Quantum and Classical Dynamics

2.1 Classical dynamics

2.1.1 Hamiltonian formulation

- Dynamics of the particles are described as trajectories in phase space with coordinates q_k and momenta p_k .
- State of system is a point in $6N$ -dimensional phase space ($3N$ coordinates and $3N$ momenta).

Hamiltonian formulation: introduced by William Rowan Hamilton (1805-1865), an Irish physicist who also introduced Hamilton's principle, Hamiltonian mechanics, Hamilton-Jacobi equation, quaternions, Nabla symbol, tensors etc.

Based on a Hamiltonian, H , the equations of motion are

$$\frac{dq_k}{dt} \equiv \dot{q}_k = \frac{\partial H}{\partial p_k} \quad (2.1a)$$

$$\frac{dp_k}{dt} \equiv \dot{p}_k = -\frac{\partial H}{\partial q_k}. \quad (2.1b)$$

For the Hamiltonian

$$H = \sum_k \frac{p_k^2}{2m_k} + V(q_1, q_2, \dots), \quad (2.2)$$

we can recover Newton's equation of motion:

$$\dot{q}_k = \frac{p_k}{m_k} \quad (2.3a)$$

$$\dot{p}_k = -\frac{dV}{dq_k} \equiv F_k. \quad (2.3b)$$

Differentiating the first line again and comparing the two shows that $\dot{p}_k = F_k = m_k \ddot{q}_k$, which is Newton's second law.

Therefore the time-evolution of a dynamical variable $A \equiv A(\vec{p}, \vec{q})$ is given by

$$\frac{dA}{dt} = \sum_k \frac{\partial A}{\partial q_k} \dot{q}_k + \frac{\partial A}{\partial p_k} \dot{p}_k \quad (2.4a)$$

$$= \sum_k \frac{\partial A}{\partial q_k} \frac{\partial H}{\partial p_k} - \frac{\partial A}{\partial p_k} \frac{\partial H}{\partial q_k} \quad (2.4b)$$

$$= \{A, H\} = -\{H, A\}, \quad (2.4c)$$

where the curly brackets are known as Poisson brackets defined as above. We can easily show that energy is conserved using the operator H in this formula to give $\frac{dH}{dt} = 0$.

2.1.2 Classical dynamics of harmonic oscillator

For a harmonic oscillator, $V(q) = \frac{1}{2}kq^2$, the equations of motion are solved by

$$q(t) = q_0 \cos \omega t + \frac{p_0}{m\omega} \sin \omega t \quad (2.5a)$$

$$p(t) = p_0 \cos \omega t - m\omega q_0 \sin \omega t, \quad (2.5b)$$

where $\omega = \sqrt{k/m}$.

2.2 Quantum dynamics

2.2.1 Propagator

The time-dependent Schrödinger equation (TDSE) is

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle. \quad (2.6)$$

This is a linear equation so the general solution of the TDSE is:

$$|\Psi(t)\rangle = \hat{U}(t, t_0) |\Psi(t_0)\rangle. \quad (2.7)$$

The time-evolution operator is obtained by solving the equation:

$$i\hbar \frac{\partial \hat{U}(t, t_0)}{\partial t} = \hat{H} \hat{U}(t, t_0). \quad (2.8)$$

The special (but common) case of a time-independent $\hat{H}$ leads to

$$\hat{U}(t, t_0) = \exp \left(-\frac{i\hat{H}(t - t_0)}{\hbar} \right), \quad (2.9)$$

where the phase factor has been included to ensure that $\hat{U}(t_0, t_0) = 1$.

Thus if $|\Psi(t_0)\rangle$ is a solution of the time-independent Schrödinger equation (TISE),

$$\hat{H} |\Psi(t_0)\rangle = E |\Psi(t_0)\rangle$$

i.e. is an eigenstate, then

$$\hat{U}(t - t_0) |\Psi(t_0)\rangle = e^{-iE(t-t_0)/\hbar} |\Psi(t_0)\rangle \quad (2.10)$$

2.2.2 Expectation values

What we have used so far is called the “**Schrödinger picture**”.

$$\begin{aligned} \text{e.g:} \quad \langle A(t) \rangle &= \langle \Psi(t) | \hat{A} | \Psi(t) \rangle \\ &= \langle e^{-\frac{i\hat{H}t}{\hbar}} \Psi(0) | \hat{A} | e^{-\frac{i\hat{H}t}{\hbar}} \Psi(0) \rangle. \end{aligned} \quad (2.11)$$

Properties:

1. Operators are stationary (time independent)
2. State vector Ψ and expectation values evolve in time.

We now turn to **The “Heisenberg picture”**. The expectation value can equivalently be written

$$\langle A(t) \rangle = \langle \Psi(0) | e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} | \Psi(0) \rangle \quad (2.12a)$$

$$= \langle \Psi(0) | \hat{A}(t) | \Psi(0) \rangle \quad (2.12b)$$

where

$$\hat{A}(t) = e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar}. \quad (2.13)$$

Properties:

1. State vector Ψ is stationary.
2. Operators are time dependent.

For the special case of time-independent Hamiltonians:

$$\frac{d}{dt} \hat{A} = \frac{i}{\hbar} \left(\hat{H} e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} - e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} \hat{H} \right) + e^{i\hat{H}t/\hbar} \frac{\partial \hat{A}}{\partial t} e^{-i\hat{H}t/\hbar} \quad (2.14a)$$

$$= \frac{i}{\hbar} [\hat{H}, \hat{A}] + \frac{\partial \hat{A}}{\partial t} \quad (2.14b)$$

The final term is 0 if the operator isn't explicitly time-dependent.

For position and momentum operators $\hat{q}$ and $\hat{p}$ and a Hamiltonian $\hat{H} = \hat{p}^2/2m + V(\hat{q})$,

$$[\hat{H}, \hat{p}] = [V(\hat{q}), \hat{p}] = i\hbar \frac{\partial V}{\partial \hat{q}} \quad (2.15a)$$

$$[\hat{H}, \hat{q}] = [\hat{p}^2/2m, \hat{q}] = -i\hbar \hat{p}/m \quad (2.15b)$$

and therefore

$$\frac{d\hat{p}}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{p}] = - \left(\frac{\partial V}{\partial \hat{q}} \right) \quad (2.16a)$$

$$\frac{d\hat{q}}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{q}] = \hat{p}/m \quad (2.16b)$$

This looks remarkably similar to the classical equations of motion.

In fact, it means that the average position/momentum follow a trajectory

$$\frac{d}{dt} \langle \hat{p}(t) \rangle = - \langle \nabla V(\hat{q}(t)) \rangle \quad \frac{d}{dt} \langle \hat{q}(t) \rangle = \langle \hat{p}(t) \rangle / m \quad (2.17a)$$

which is known as the Ehrenfest theorem. In special cases only, $\langle \nabla V(\hat{q}) \rangle = \nabla V(\langle \hat{q} \rangle)$, e.g. for the harmonic oscillator where the gradient is a linear function of the position, and the average position/momentum really do exactly follow the ordinary classical trajectory. For this reason, the dynamics of the quantum harmonic oscillator has many similarities to the dynamics of the classical harmonic oscillator. However, in general, the average of the wavepacket only approximately follows the classical motion and can deviate significantly after a long propagation time.

2.2.3 Differences between mixed states and superpositions of states

A linear combination (or superposition) of eigenstates such as

$$\Psi(x) = c_1\psi_1(x) + c_2\psi_2(x) \quad (2.18)$$

depends on time as

$$\Psi(x, t) = c_1\psi_1(x, t) + c_2\psi_2(x, t) \quad (2.19a)$$

$$= c_1e^{-iE_1t/\hbar}\psi_1(x) + c_2e^{-iE_2t/\hbar}\psi_2(x). \quad (2.19b)$$

Thus the probability density oscillates as

$$|\Psi(x, t)|^2 = |c_1|^2|\psi_1(x)|^2 + |c_2|^2|\psi_2(x)|^2 + 2\operatorname{Re}[c_1^*c_2\psi_1(x)^*\psi_2(x)]\cos(E_2 - E_1)t/\hbar. \quad (2.20)$$

The dynamics of mixed states is simple as they are independent. For the density operator,

$$\hat{\rho} = P_1|\psi_1\rangle\langle\psi_1| + P_2|\psi_2\rangle\langle\psi_2|, \quad (2.21)$$

there is no time dependence,

$$\hat{\rho}(t) = e^{-iE_1t/\hbar}P_1e^{iE_1t/\hbar}|\psi_1\rangle\langle\psi_1| + e^{-iE_2t/\hbar}P_2e^{iE_2t/\hbar}|\psi_2\rangle\langle\psi_2| = \hat{\rho}.$$

This is true only for density matrices which are diagonal in the eigenstate basis, and the interference terms appear as a consequence of off-diagonal elements. The density matrix

$$\hat{\rho} = c_1c_1^*|\psi_1\rangle\langle\psi_1| + c_1c_2^*|\psi_1\rangle\langle\psi_2| + c_2c_1^*|\psi_2\rangle\langle\psi_1| + c_2c_2^*|\psi_2\rangle\langle\psi_2| \quad (2.22)$$

has the time dependence

$$\hat{\rho}(t) = |c_1|^2|\psi_1\rangle\langle\psi_1| + e^{-i(E_1-E_2)t/\hbar}c_1c_2^*|\psi_1\rangle\langle\psi_2| + e^{-i(E_2-E_1)t/\hbar}c_2c_1^*|\psi_2\rangle\langle\psi_1| + |c_2|^2|\psi_2\rangle\langle\psi_2|. \quad (2.23)$$

The off-diagonal elements are thus termed “coherence” terms.

2.2.4 Tunnelling in ammonia

We first consider states in the localized left or right potential minimum ϕ_ℓ , ϕ_r . The lowest lying stationary states have energies E_0 and E_1 . The corresponding wave functions are

$$\psi_0 \simeq \frac{1}{\sqrt{2}}(\phi_\ell + \phi_r) \quad (2.24a)$$

$$\psi_1 \simeq \frac{1}{\sqrt{2}}(\phi_\ell - \phi_r). \quad (2.24b)$$

We assume that at $t = 0$ the wave packet is localized in the left well.

$$\Psi(t = 0) = \frac{1}{\sqrt{2}}(\psi_0 + \psi_1) = \phi_\ell \quad (2.25)$$

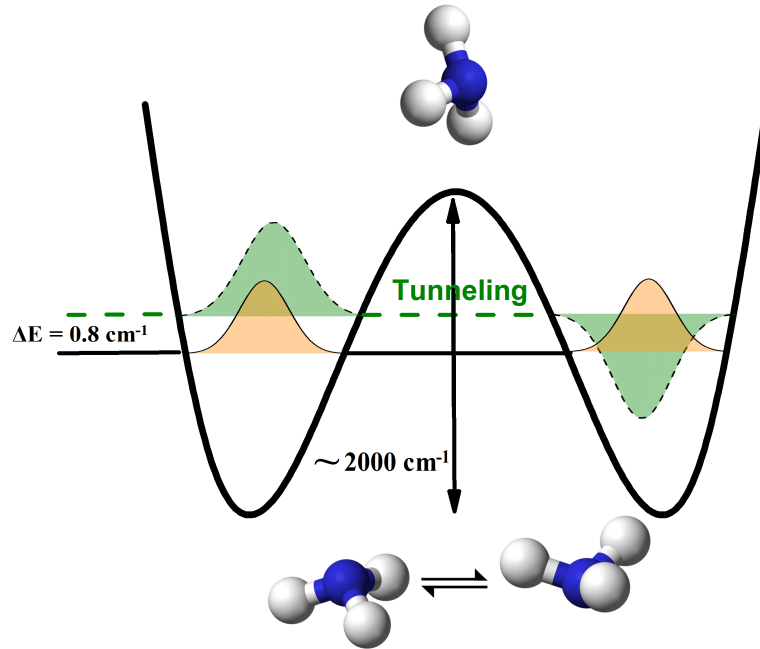


Figure 1: Tunnelling and ammonia inversion

This is not an eigenstate (stationary solution) but a superposition state or wave packet.

$$\Psi(t) = \frac{1}{\sqrt{2}} e^{\frac{-iE_0 t}{\hbar}} (\psi_0 + \psi_1 e^{\frac{-i(E_1 - E_0)t}{\hbar}}). \quad (2.26)$$

For $t_1 = \frac{\hbar}{2(E_1 - E_0)}$ the wave function becomes

$$\Psi(t_1) = \frac{1}{\sqrt{2}} (\psi_0 - \psi_1) = \phi_r \quad (2.27)$$

which gives the oscillation period $T = \frac{\hbar}{\Delta E}$ where $\Delta E = E_1 - E_0$ is the tunnelling splitting. This relation is a general result for 2-state wave packet.

The wavefunction returns to its initial state, called a revival.

2.3 Observables

2.3.1 Time averages

The probability density in classical mechanics is $\rho(\vec{p}, \vec{q})$, for instance $\rho_{\text{eq}} = e^{-\beta H(\vec{p}, \vec{q})}/Q$ is the Boltzmann distribution. The average of an observable $A \equiv A(\vec{p}, \vec{q})$ in a particular distribution ρ is

$$\langle A(t) \rangle_{\rho} = (2\pi\hbar)^{-f} \iint \rho(\vec{p}_0, \vec{q}_0) A(\vec{p}_t, \vec{q}_t) d\vec{p}_0 d\vec{q}_0. \quad (2.28)$$

For the special case of the the equilibrium distribution, which due to conservation of energy, is time independent,

$$\langle A(t) \rangle_{\text{eq}} = (2\pi\hbar)^{-f} \iint A(\vec{p}_t, \vec{q}_t) \rho_{\text{eq}}(\vec{p}_t, \vec{q}_t) d\vec{p}_0 d\vec{q}_0 \quad (2.29)$$

Liouville's theorem states that $d\vec{p}_0 d\vec{q}_0 = d\vec{p}_t d\vec{q}_t$ such that

$$\langle A(t) \rangle_{\text{eq}} = (2\pi\hbar)^{-f} \iint A(\vec{p}_t, \vec{q}_t) \rho_{\text{eq}}(\vec{p}_t, \vec{q}_t) d\vec{p}_t d\vec{q}_t \quad (2.30a)$$

$$= (2\pi\hbar)^{-f} \iint A(\vec{p}, \vec{q}) \rho_{\text{eq}}(\vec{p}, \vec{q}) d\vec{p} d\vec{q} \quad (2.30b)$$

$$= \langle A(0) \rangle_{\text{eq}} \equiv \langle A \rangle_{\text{eq}} \quad (2.30c)$$

where we simply rename the integration variables to prove equivalence with the average value of A at time 0.

So although properties such as bond lengths are oscillating, the ensemble averaged bond length remains constant when in equilibrium.

With quantum mechanics, similar formulae exist

$$\langle \hat{A}(t) \rangle_{\rho} = \text{Tr}[\hat{\rho} \hat{A}(t)] \quad (2.31a)$$

$$= \text{Tr} \left[\hat{\rho} e^{i\hat{H}t/\hbar} \hat{A} e^{-i\hat{H}t/\hbar} \right] \quad (2.31b)$$

$$= \text{Tr} \left[e^{-i\hat{H}t/\hbar} \hat{\rho} e^{i\hat{H}t/\hbar} \hat{A} \right] \quad (2.31c)$$

$$= \text{Tr} \left[\hat{\rho}(t) \hat{A} \right] \quad (2.31d)$$

where

$$\hat{\rho}(t) = e^{-i\hat{H}t/\hbar} \hat{\rho} e^{i\hat{H}t/\hbar}, \quad (2.32)$$

which implies $\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}]$ (compare with equation (2.13) equation (2.14b) and note minus sign difference). For the equilibrium case,

$$\langle \hat{A}(t) \rangle_{\text{eq}} = \text{Tr} \left[\hat{\rho}_{\text{eq}}(t) \hat{A} \right] \quad (2.33a)$$

$$= \text{Tr} \left[\hat{\rho}_{\text{eq}} \hat{A} \right] \equiv \langle \hat{A} \rangle_{\text{eq}} \quad (2.33b)$$

as the density operator is time independent, i.e. commutes with propagators.

2.3.2 Correlation functions

In order to gain information about the dynamical processes, we define time-correlation functions.

$$\langle A(0)B(t) \rangle = (2\pi\hbar)^{-f} \iint \rho(\vec{p}_0, \vec{q}_0) A(\vec{p}_0, \vec{q}_0) B(\vec{p}_t, \vec{q}_t) d\vec{p}_0 d\vec{q}_0 \quad (2.34)$$

After a long time the correlation between two measurements decays, i.e. the system decorrelates and $\langle A(0)B(t) \rangle \rightarrow \langle A \rangle \langle B \rangle$ as $t \rightarrow \infty$. This is a consequence of chaotic dynamics.

Quantum case

$$\langle \hat{A}(0)\hat{B}(t) \rangle = \text{Tr} \left[\hat{\rho} \hat{A} e^{i\hat{H}t/\hbar} \hat{B} e^{-i\hat{H}t/\hbar} \right] \quad (2.35)$$

2.3.3 Example: harmonic oscillator

To compute classical position autocorrelation functions for harmonic oscillator, use $A = B = x$.

$$\langle x(0)x(t) \rangle = \frac{1}{2\pi\hbar} \iint \rho_{\text{eq}}(x_0, p_0) x_0 x_t dx_0 dp_0 \quad (2.36a)$$

$$= \frac{1}{2\pi\hbar Q} \iint e^{-\beta H(x_0, p_0)} x_0 x_t dx_0 dp_0 \quad (2.36b)$$

$$= \frac{\beta\hbar\omega}{2\pi\hbar} \iint e^{-\beta \frac{p_0^2}{2m} - \frac{1}{2}\beta m\omega^2 x_0^2} x_0 \left[x_0 \cos \omega t + \frac{p_0}{m\omega} \sin \omega t \right] dx_0 dp_0 \quad (2.36c)$$

$$= \frac{\beta\omega}{2\pi} \sqrt{\frac{2\pi m}{\beta}} \int e^{-\frac{1}{2}\beta m\omega^2 x_0^2} x_0^2 \cos \omega t dx_0 \quad (2.36d)$$

$$= \frac{\beta\omega}{2\pi} \sqrt{\frac{2\pi m}{\beta}} \sqrt{\frac{2\pi}{(\beta m\omega^2)^3}} \cos \omega t \quad (2.36e)$$

$$= \frac{1}{\beta m\omega^2} \cos \omega t \quad (2.36f)$$

2.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Fundamentals of Quantum mechanics

- Dirac, Quantum Mechanics (Oxford, 1948)

Time-dependent Quantum mechanics

- Tannor, Introduction to quantum mechanics: a time-dependent perspective

Time-dependent ensembles

- Tuckerman, Statistical Mechanics (Oxford)
- Zwanzig, Nonequilibrium statistical mechanics (Oxford)

Properties of correlation functions

- Nitzan, Chemical Dynamics in Condensed Phases (Oxford) Sec. 6.3

Molecular Dynamics

- Tuckerman, Statistical Mechanics (Oxford)
- Frenkel and Smit, Understanding Molecular Simulations (Academic Press)